Experiment-7

Implement Packet Sniffing using Raw Sockets in Python

Aim:

To implement a Packet Sniffing using Raw sockets in Python capture and analyze network packets.

Procedure:

1. open a terminal with root Privilage
2. create a Python Script 'sniffer.py'
3. import socket, struct, textwrap and binascii modules.
4. create a raw socket using socket AF_Packet (Linux) or socket AF_INET with socket.
5. Bind the socket to the network interface.
6. use a loop to continously receive Packet with socket.recvfrom().
7. unique the Element header using struct.unpack().

Command/code:

```
# Packet-snitter.py
import socket, struct, binasc

def mac-addr(raw):
    return ':'.join('%02x' for b in raw)

s = socket.socket(socket.AF_PACKET,
    socket.SOCK_RAW, ntohs(0x0003))
while True:
    raw_addr = s.recvfrom(65535)
    eth = struct.unpack('!6s6sH', raw)
    dst = mac-addr(eth[0]); src = mac-addr(eth[1])
    print('src: {src} dst: {dst}
    Proto {Proto: #06x}')
```

8. Extract and Print the Source mac address, destination mac, and Protocol.

9. For IP Packets (Ether Type 0x800), unpack the IP header.

10. Display Source IP, destination IP, TTL, and the Protocol number.

11. Run the script and observe captured packets.

Results

The Packet Sniffer was successfully implemented. Ethernet frames and IP packets were captured and source destination mac and IP address along with Protocol Information were displayed